PrimeTek Africa

Codecamp Academy

NoSQL Databases



Presented by:
Cristian Camilo Cortes Ortiz

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NoSQL Databases

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Document-based

Stores data as flexible documents (e.g., JSON) instead of rows.

Each key maps to a document containing nested fields, arrays, and varying structures.

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Cloud Firestore



Couchbase

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Graph-based

Stores data as nodes and edges, optimized for relationships.

Ideal for modeling networks social connections, recommendations, fraud detection.

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Key-Value

Simplest NoSQL model: data stored as key-value pairs.

Fast lookups, no fixed schema, the value can be anything from a string to a JSON blob.

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Column-Oriented

Stores data in columns grouped into column families, rather than rows.

Optimized for fast reads/writes on large datasets and analytical queries.

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What advantages does NoSQL offer in terms of scalability?

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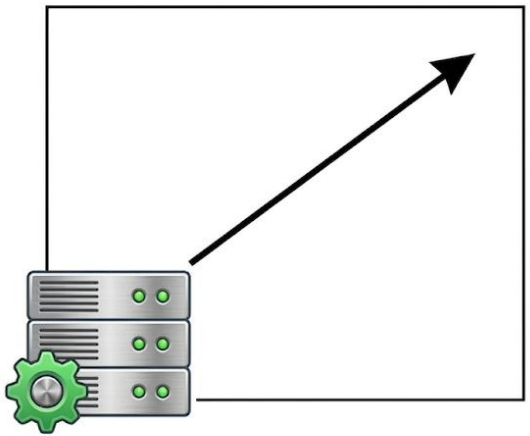
One of the major advantages of NoSQL databases is their ability to scale effectively.
Scaling is a critical process for maintaining and improving database performance as the volume of data and the number of users increase.

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Vertical scaling

This involves increasing the capacity of a single server by adding more resources such as CPU, RAM, or storage. Although it is a simple solution, it can be costly in the long run.

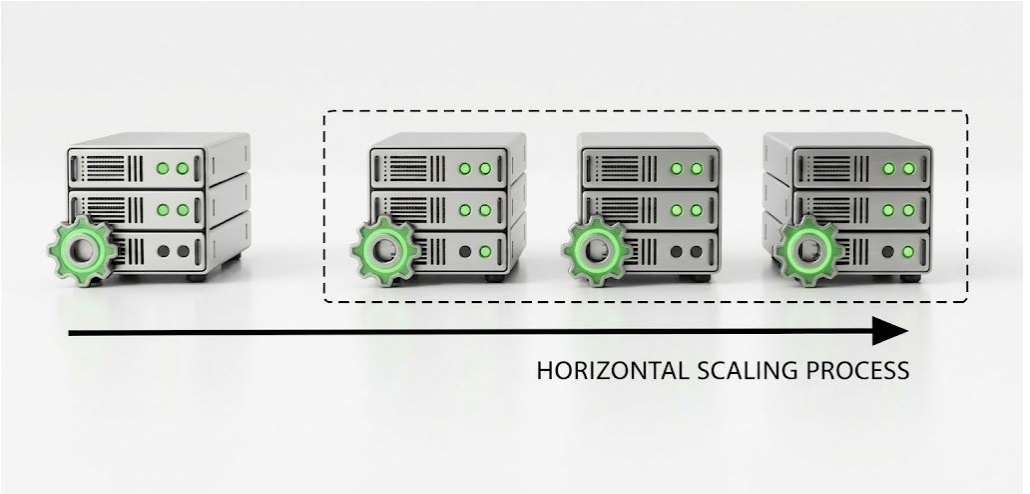


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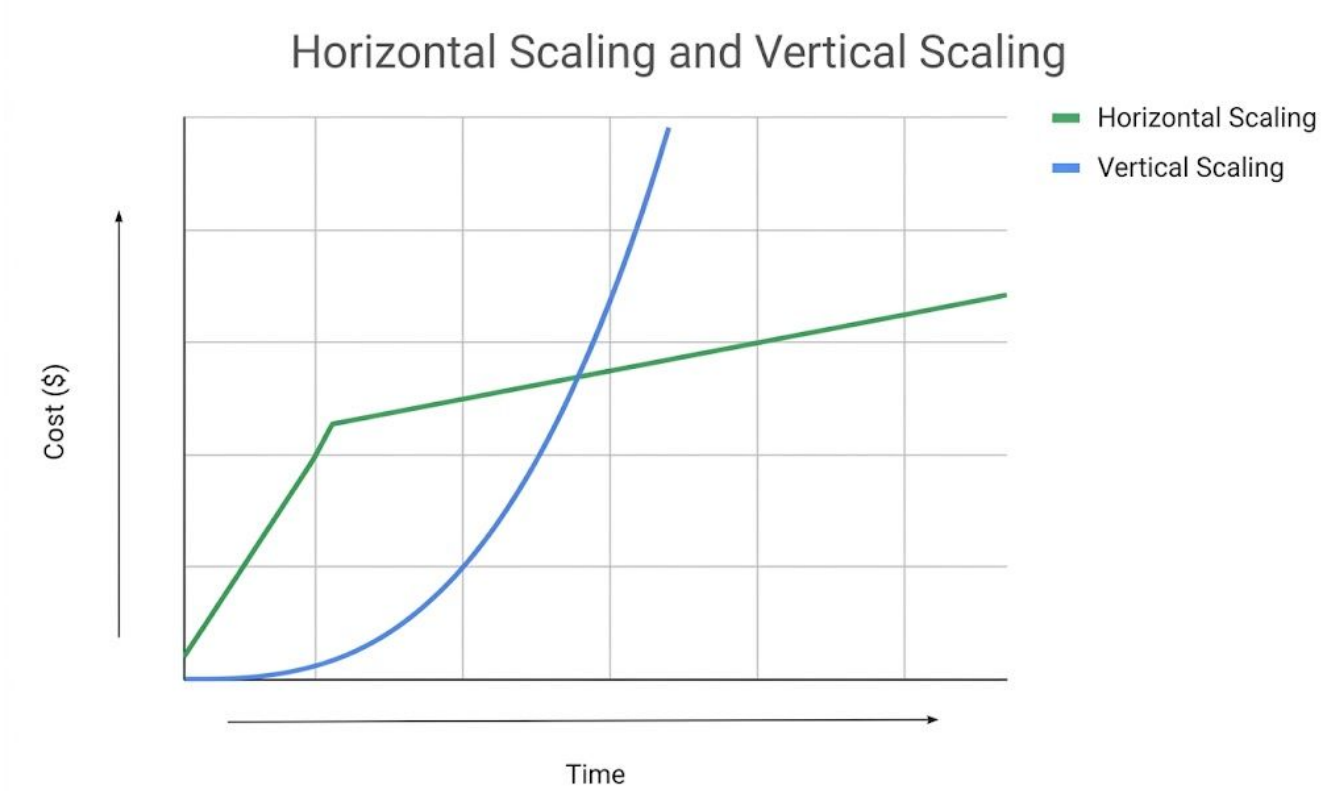
Horizontal scaling

It involves duplicating nodes or servers to handle the workload in a distributed manner. NoSQL maximizes this technique through data replication, ensuring high availability and fault tolerance.



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How does replication work in NoSQL databases?

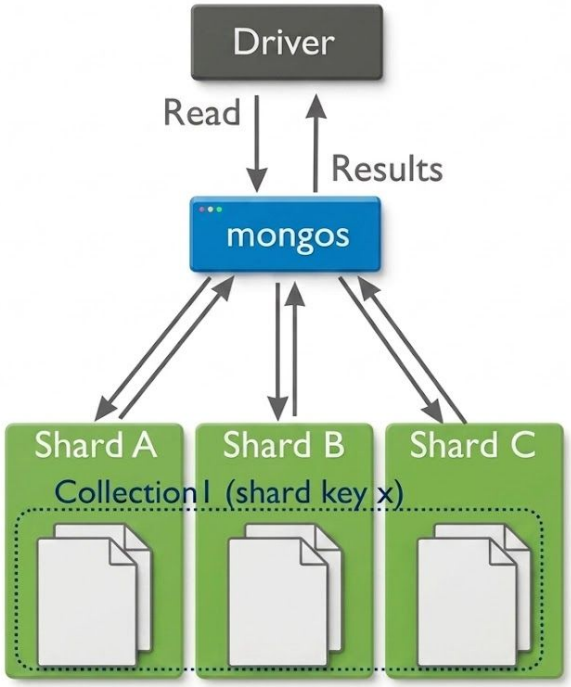
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Replication is a key process in NoSQL databases for ensuring high availability and fault tolerance, and it enables parallel processing of requests. It is managed by a load balancer, which directs queries to available nodes. In the event of a failure, another node takes over the task without affecting service.

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What are documents?

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Documents are an essential part of document databases such as MongoDB. These documents store domain-specific information, such as the products in an online store or the inventory of a brick-and-mortar store. A document groups a set of related data into key-value pairs, providing flexibility and efficiency in data storage.

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```
{
  "_id": "5c8eccc1caa187d17ca6ed16",
  "city": "ALPINE",
  "zip": "35014",
  "loc": {
    "y": 33.331165,
    "x": 86.208934
  },
  "pop": 3062,
  "state": "AL"
}
```


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```
{
  "_id": "5c8eccc1caa187d17ca6ed16",
  "city": "ALPINE",
  "zip": "35014",
  "loc": {
    "y": 33.331165,
    "x": 86.208934
  },
  "pop": 3062,
  "state": "AL"
}
```


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```
{
  "_id": "5c8eccc1caa187d17ca6ed16",
  "city": "ALPINE",
  "zip": "35014",
  "loc": {
    "y": 33.331165,
    "x": 86.208934
  },
  "pop": 3062,
  "state": "AL"
}
```


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What are collections?

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Collections are groups of documents that share a certain relationship or data model. In MongoDB and other document databases such as Firestore and Couchbase, collections are the equivalent of tables in a relational database, but without rigid schema constraints.

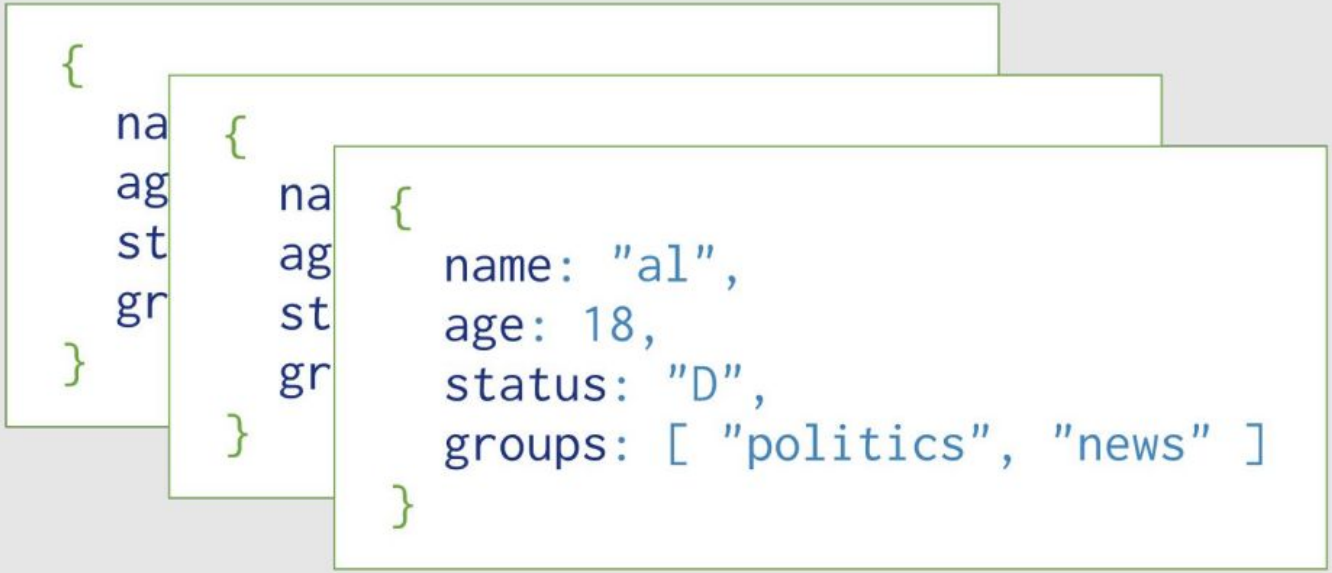
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```
{
  name: "al",
  age: 18,
  status: "D",
  groups: [ "politics", "news" ]
}
```

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What is Mongosh?

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A command-line interface for interacting with MongoDB. Lets you query, insert, update, and manage databases directly using JavaScript syntax.

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What is Mongo Compass?

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The official GUI for MongoDB. A visual tool for exploring and managing data. Lets you view documents, run queries, analyze schemas, and build aggregations without writing code.

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Functionality

It allows you to perform searches, manage databases, and view detailed information about collections and documents.

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Compatibility

Available for various operating systems, including Windows, Mac, and Ubuntu.

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What are the benefits of Mongo Compass?

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Visual interface

It makes it easy to interact with the database without having to write complex code.

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Connection Management

It lets you save frequently visited sites as favorites for quick access.

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Interactive Queries

It offers a simple way to run queries, view results, and manage data.

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What is Mongo Atlas?

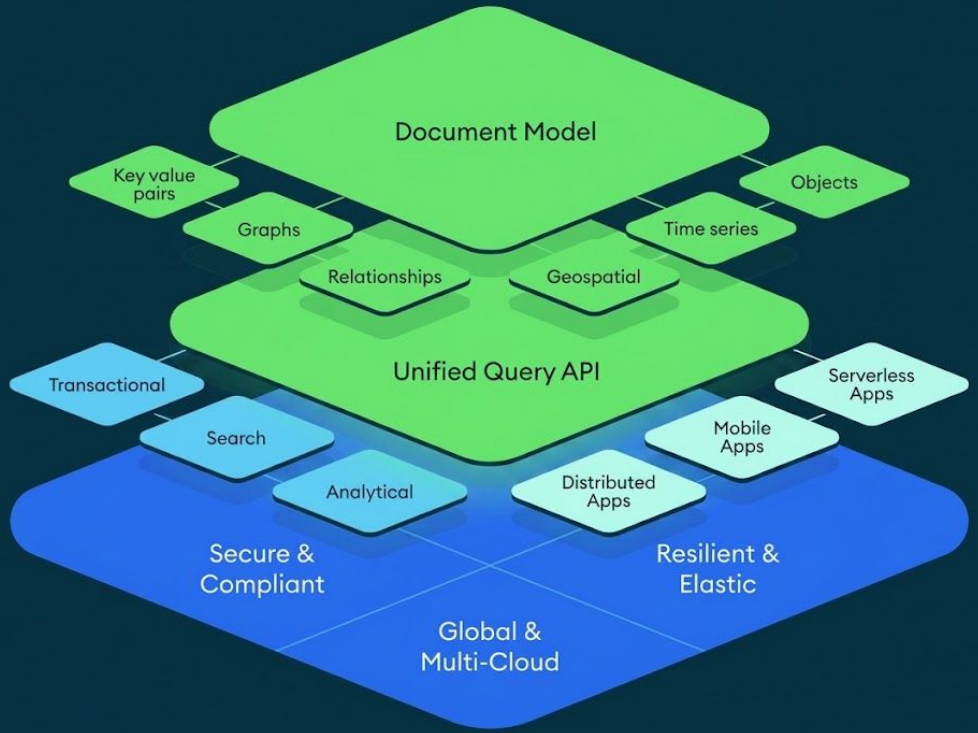
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MongoDB Atlas is a cloud service that offers effective solutions for managing MongoDB databases. Its cloud architecture reduces the complexity of replication and clustering, two key aspects of database administration.

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Preconfigured clustering

It enables a model for replication and functional scalability with just a couple of clicks.

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Free Plans

They offer up to 5 GB of storage, shared RAM, and usage alerts without requiring a credit card.

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Loading sample datasets

These allow users to run queries and manage large volumes of data, including well-known examples such as Airbnb's databases.

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Free Cluster

- 512MB to 5GB of storage.
- Shared RAM.
- Upgrade to dedicated clusters for full functionality.
- No credit card required to start.

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```
// Code to create a user and define IP
// Addresses for accessing the cluster
db.createUser({
  user: "CarlAdmin",
  pwd: "CarlAdmin123",
  roles: [{ role: "readWrite", db: "admin" }]
})

// Set Access IP
{
  "ip": "0.0.0.0/0",
  "comment": "localhost"
}
```

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MongoDB in VSCode

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An official extension that brings MongoDB directly into your editor. Connect to databases, browse collections, and run queries without leaving VS Code.

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MongoDB for VS Code

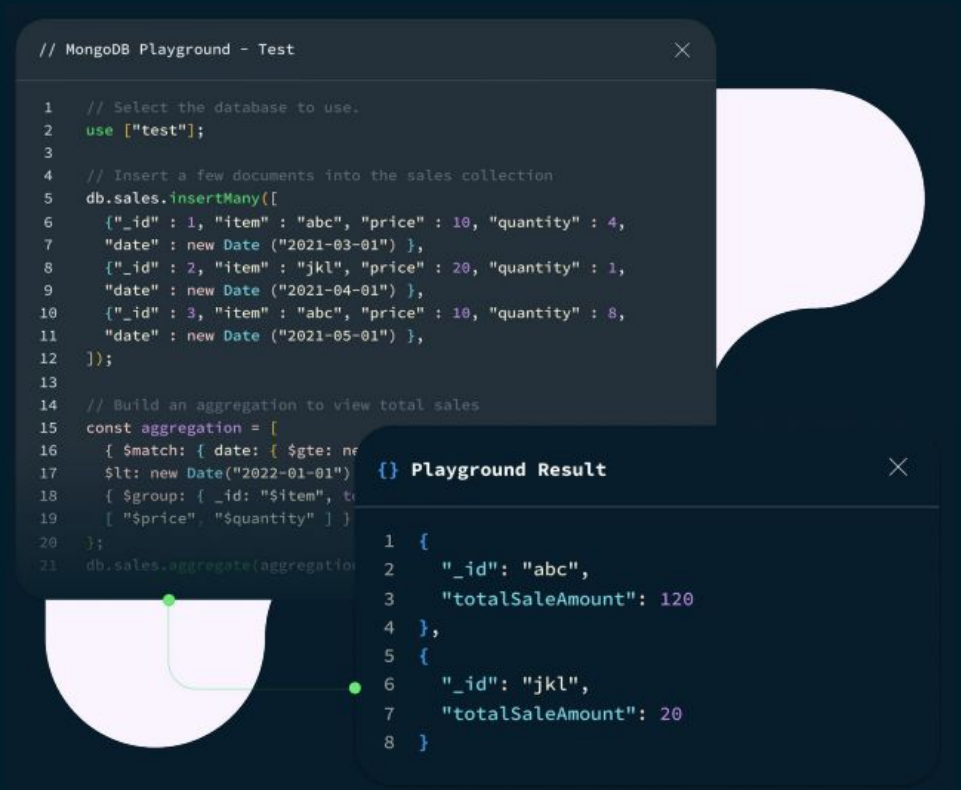
MongoDB mongodb.com | 3,160,538 | (43)

Connect to MongoDB and Atlas directly from your VS Code environment, navigate your databases and collections, inspect your schema and use playgrounds to prototype queries and aggregations.

Install | Auto Update

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MongoDB Running in Docker

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Run MongoDB in an isolated container; no local installation needed. Quick to spin up, easy to tear down, and consistent across dev environments.

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Why use Docker to run MongoDB?

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Docker is a system that allows you to run any software inside a container an isolated part of your computer that operates efficiently and conveniently.

If you're working with MongoDB today and want to try Cassandra or CouchDB tomorrow, you don't need to accumulate drivers: just spin up a container, run your test, and delete it without any impact on your operating system.

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JSON vs. BSON

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¿Where are the differences between JSON and BSON?

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In the world of data generation, choose the appropriate format for this purpose to optimize performance and compatibility. JSON and BSON son of the formats widely discussed in this area. Understanding the differences on the ground will help you choose what is best for your needs and also understand how MongoDB manages information internally.

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Advantages of JSON

- friendly.
- It's readable.
- It's a widely used format.

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Disadvantages of JSON

- Text-based.
- Takes up a lot of space.
- Has limited support: strings, booleans, numbers, arrays.

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Advantages of BSON

- Binary representation of JSON.
- Takes up no space.
- High performance.
- Data types: +, date, raw binary, integer, long, float.

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Disadvantages of BSON

- It is not standard.
- It is a machine-readable format.

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```
{
    "hello" : "world"
}
```

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```
\x16\x00\x00\x00                  // total document size
\x02                              // 0x02 = type String
hello\x00                         // field name
\x06\x00\x00\x00world\x00         // field value
\x00                              // 0x00 = end of object
```

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Operators

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Updating a document

Operators

- **$set**
  Sets the value of a field in a document.
- **$inc**
  Increments the value of the field by the specified amount.
- **ObjectId**
  Function to find a doc with objID.

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Updating several document

Operators

- **$set**
  Sets or create the value of a field in a document.
- **$rename**
  Renames a field.
- **$unset**
  Removes the specified field from a document.

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Array Update Operators

Operators

- **$push**
  Adds an item to an array.
- **$pull**
  Removes all array elements that match a specified query.
- **$in**
  Matches any of the values specified in an array.

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Insert or Update

Operators

- **$pop**
  Removes the first or last item of an array.

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Comparison Query Operators

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Operators

- **$eq**
  Matches values that are equal to a specified value.
- **$ne**
  Matches all values that are not equal to a specified value.

Comparison Query Operators

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Operators

- **$gte**
  Matches values that are greater than or equal to a specified value.
- **$gt**
  Matches values that are greater than a specified value.

Comparison Query Operators

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- **$lte**
  Matches values that are less than or equal to a specified value.
- **$lt**
  Matches values that are less than a specified value.

Regular Expressions

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Operators

- **$regex**
  Allows use of aggregation expressions within the query language.

Projection

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Array Operators

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- **$in**
  Matches any of the values specified in an array.
- **$nin**
  Matches none of the values specified in an array.

Array Operators

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Operators

- **$all**
  Matches arrays that contain all elements specified in the query.
- **$elemMatch**
  Selects documents if element in the array field matches all the specified $elemMatch conditions.

Array Operators

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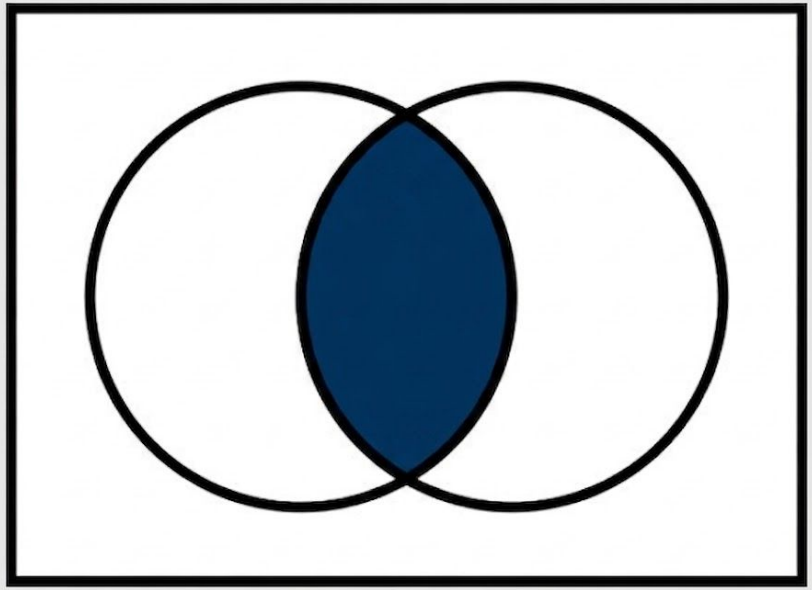
- **$size**
  Selects documents if the array field is a specified size.

Logical Operators

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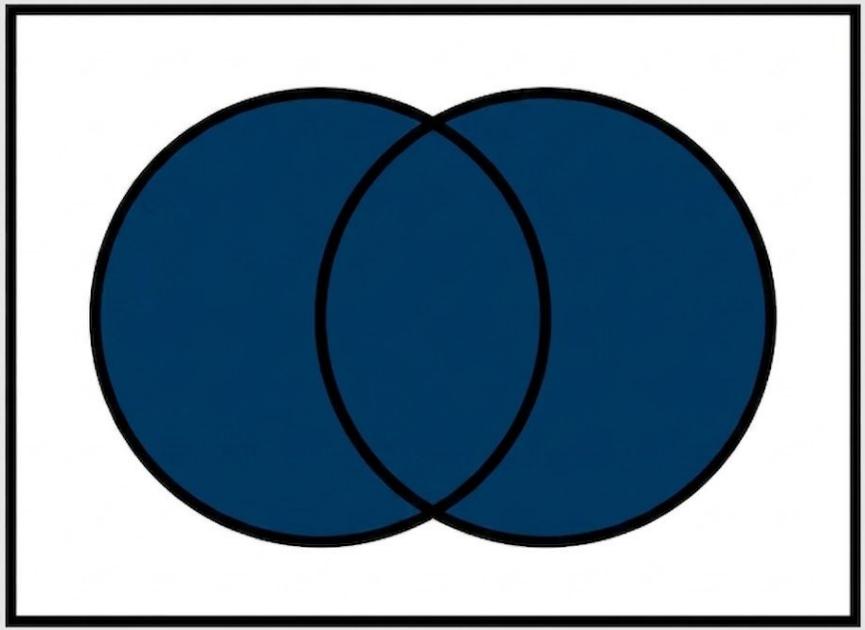
AND



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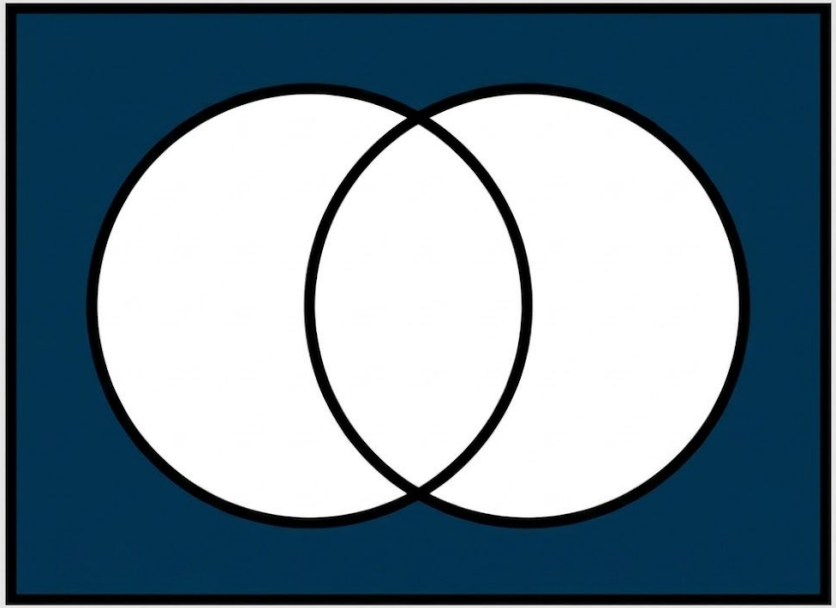
OR



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NOR



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Logical Operators

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Operators

- **$and**
  Joins query clauses with a logical AND returns all documents that match the conditions of both clauses.

- **$not**
  Inverts the effect of a query expression and returns documents that do not match the query expression.

Logical Operators

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Operators

- **$nor**
  Joins query clauses with a logical NOR returns all documents that fail to match both clauses.

- **$or**
  Joins query clauses with a logical OR returns all documents that match the conditions of either clause.

Expressive operator

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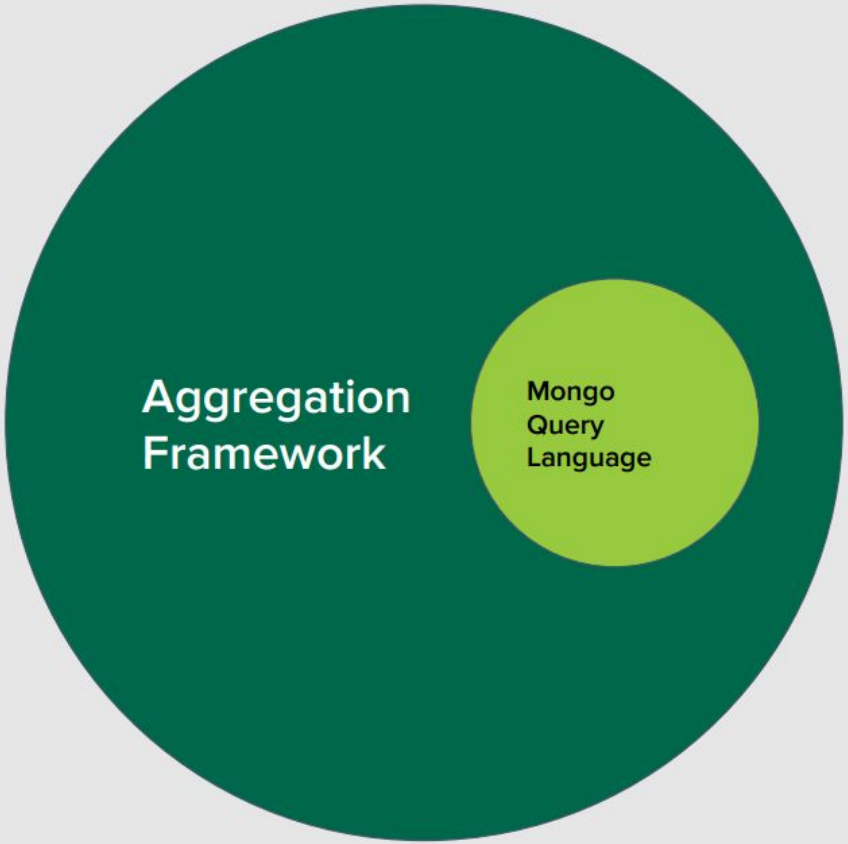
Operators

- **$expr**
  Allows use of aggregation expressions within the query language.

Aggregation Framework

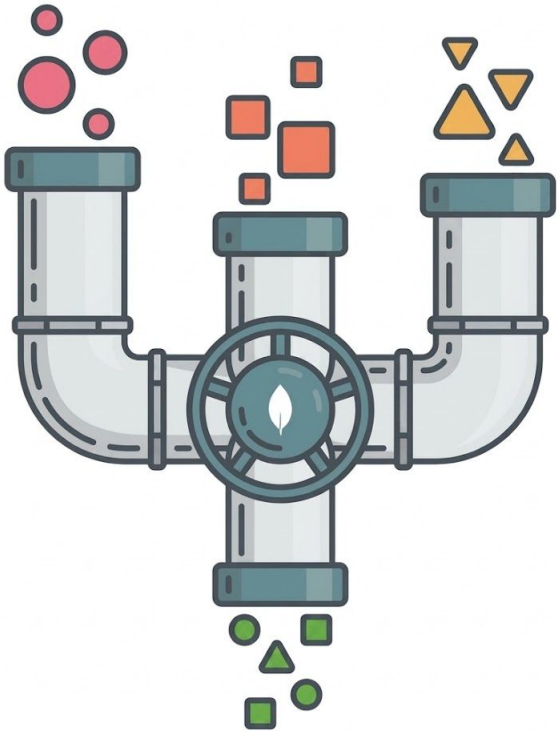
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Sort & Limit

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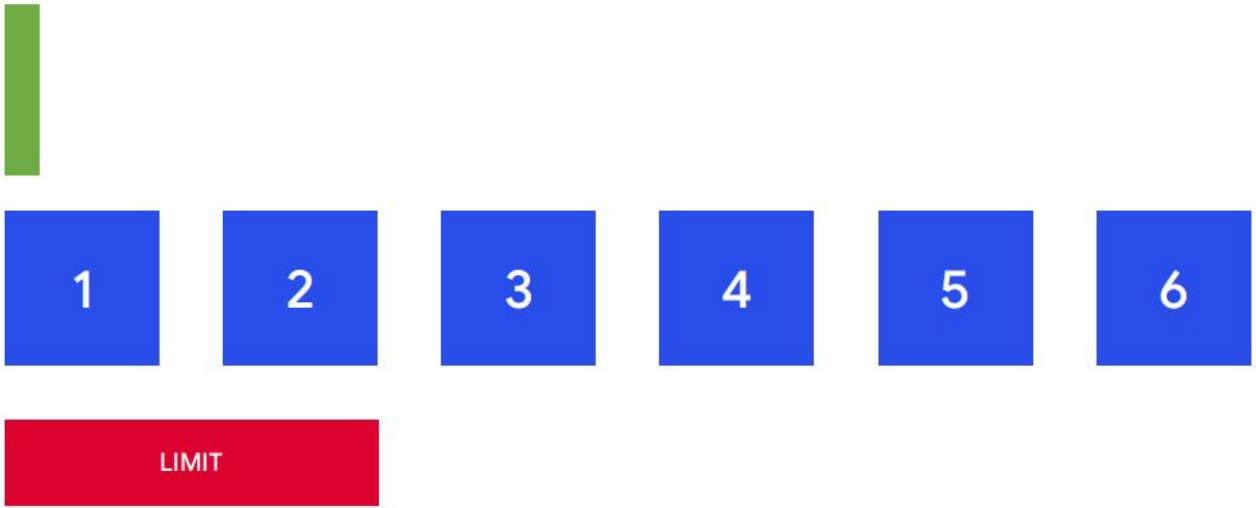
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Pagination

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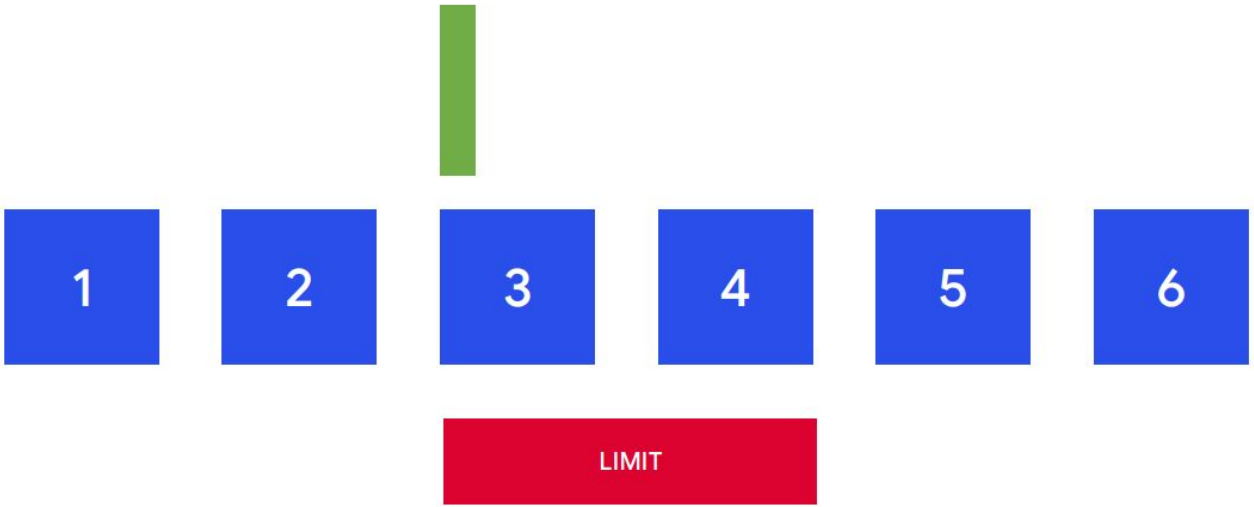
limit = 2
skip / offset = 0



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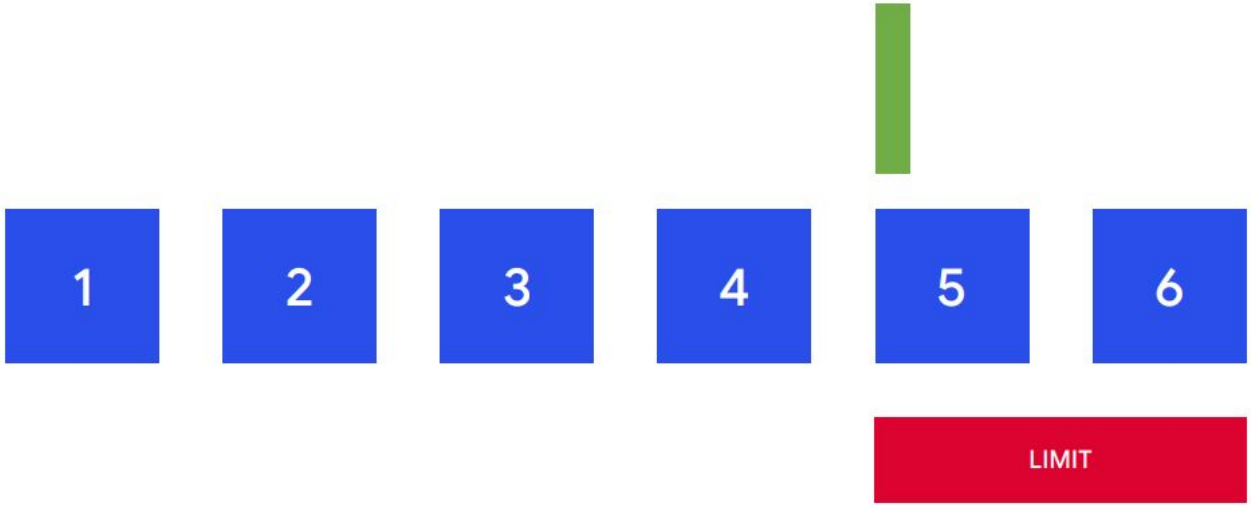
limit = 2
skip / offset = 2



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limit = 2
skip / offset = 4



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Connecting to MongoDB

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Drivers
Client Libraries

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```
from pymongo import MongoClient

client = MongoClient(port=27017)

db=client.business

fivestarcount = db.reviews.find({'rating': 5}).count()

result = db.reviews.update_one({...})
```

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```
const { MongoClient } = require("mongodb");

const client = new MongoClient("uri");

const database = client.db('sample_mflix');

const movies = database.collection('movies');

const query = { title: 'Back to the Future' };
const movie = await movies.findOne(query);
```

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You've built a solid MongoDB foundation; keep querying, modeling, and creating.

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